

复现 Nature Communications 图 4f: Hep G2 剂量-反应曲线

ent-Paroxol 与 3-Fluoro-5-hydroxybenzonitrile 的细胞毒性

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1 背景

本文复现 Chen 等 (*Nat. Commun.* 2024, 15:8396) 图 4f: 用 CCK-8 法测得的 Hep G2 细胞在两种候选污染物不同浓度下的存活率 (cell viability), 并拟合四参数对数逻辑斯蒂 (4PL) 剂量-反应曲线以估计 IC_{50} 。原文报告 ent-Paroxol 的 IC_{50} 为 494.4 μM 、3-Fluoro-5-hydroxybenzonitrile 为 545.9 μM , 每个浓度 6 个生物学重复, 误差棒为标准差。

```
library(tidyverse)
library(minpack.lm) # nlsLM: Levenberg-Marquardt, 比 base nls 更稳健
library(broom)
library(cmdstanr)
library(posterior)
library(showtext)
showtext_auto()

compound_levels <- c("ent-Paroxol", "3-Fluoro-5-hydroxybenzonitrile")

raw_data <- read_csv("data/fig4f.csv") |>
  mutate(compound = fct_relevel(compound, compound_levels))

head(raw_data) |>
  knitr::kable(digits = 3)
```

compound	conc_um	replicate	viability
3-Fluoro-5-hydroxybenzonitrile	0	1	1
3-Fluoro-5-hydroxybenzonitrile	0	2	1
3-Fluoro-5-hydroxybenzonitrile	0	3	1
3-Fluoro-5-hydroxybenzonitrile	0	4	1
3-Fluoro-5-hydroxybenzonitrile	0	5	1
3-Fluoro-5-hydroxybenzonitrile	0	6	1

原始数据为长表: **compound** (化合物)、**conc_um** (浓度 μM)、**replicate** (重复编号)、**viability** (存活率)。把化合物设为有序因子, 使图例顺序与配色和原文一致 (ent-Paroxol 在前)。

2 汇总为 mean $\pm$ sd

绘制的点与误差棒来自每个”化合物 $\times$ 浓度”组合的均值与标准差。**conc_um == 0** 是归一化对照 (存活率恒为 1), 无法置于对数横轴, 仅在建模时保留、绘图时滤除。

```
summary_data <- raw_data |>
  filter(conc_um > 0) |>
  group_by(compound, conc_um) |>
  summarise(
    mean = mean(viability),
    sd   = sd(viability),
    n    = n(),
    .groups = "drop"
  )

summary_data |>
  knitr::kable(digits = 3)
```

compound	conc_um	mean	sd	n
ent-Paroxol	5	0.964	0.094	6
ent-Paroxol	10	0.838	0.098	6
ent-Paroxol	50	0.846	0.075	6
ent-Paroxol	100	0.801	0.105	6
ent-Paroxol	500	0.446	0.031	6
ent-Paroxol	1000	0.020	0.005	6
ent-Paroxol	1500	0.002	0.001	6
ent-Paroxol	2000	0.004	0.002	6
3-Fluoro-5-hydroxybenzotrile	5	1.007	0.025	6
3-Fluoro-5-hydroxybenzotrile	10	0.907	0.060	6
3-Fluoro-5-hydroxybenzotrile	50	0.927	0.069	6
3-Fluoro-5-hydroxybenzotrile	100	0.954	0.102	6
3-Fluoro-5-hydroxybenzotrile	500	0.496	0.048	6
3-Fluoro-5-hydroxybenzotrile	1000	0.176	0.032	6
3-Fluoro-5-hydroxybenzotrile	1500	0.060	0.017	6
3-Fluoro-5-hydroxybenzotrile	2000	0.007	0.001	6

3 拟合 4PL 剂量-反应曲线

采用四参数对数逻辑斯蒂模型

$$f(x) = \text{bottom} + \frac{\text{top} - \text{bottom}}{1 + (x/\text{IC}_{50})^{\text{hill}}}$$

其中 **top**/**bottom** 为上/下平台, **IC50** 为半数抑制浓度, **hill** 为斜率。

3.1 对数逻辑斯蒂: logistic 体现在哪里

把模型里的 Hill 项改写成指数形式, 令

$$(x/\text{IC}_{50})^{\text{hill}} = e^z$$

则模型变为

$$f(x) = \text{bottom} + (\text{top} - \text{bottom}) \cdot \frac{1}{1 + e^z}$$

这里的 z , 可以通过对 $(x/\text{IC}_{50})^{\text{hill}} = e^z$ 两边取对数得到

$$z = \text{hill} \cdot \ln \frac{x}{\text{IC}_{50}} = \text{hill} (\ln x - \ln \text{IC}_{50})$$

所以, 数学表达式可以重新写为

$$y = \text{bottom} + (\text{top} - \text{bottom}) p$$

$$p = \frac{1}{1 + e^z}, \quad z = \text{hill} (\ln x - \ln \text{IC}_{50})$$

当然, 也可以写为更好理解的形式

$$y = \text{bottom} + (\text{top} - \text{bottom}) p$$

$$\text{logit}(p) = \ln \frac{p}{1 - p} = \text{hill} (\ln \text{IC}_{50} - \ln x)$$

“对数”指剂量走 $\ln x$, “逻辑斯蒂”指 p 与 $\ln x$ 之间用 logit 相连。

3.2 变量含义和取值范围

- **bottom**: 下渐近线, 即浓度趋于无穷大时的响应 (viability)。本数据中表示高剂量下的最低细胞活力, 可以取负值 (NLS 里 ent-Paroxol 为 -0.158)。取值范围 $(-\infty, \text{top})$, 拟合时约束在 $[-5, 1]$ 。
- **top**: 上渐近线, 即浓度为 0 (对照/未处理) 时的响应。CCK-8 归一化后约等于 1。取值范围 (bottom, ∞) , 拟合时约束在 $[0, 3]$ 。
- **IC50**: 半数效应浓度, 即响应恰好落在 **top** 与 **bottom** 正中间 $((\text{top} + \text{bottom})/2)$ 时所对应的浓度, 单位 μM 。恒为正, 取值范围 $(0, \infty)$, 拟合时约束在 $[1, 10^5]$ 。
- **hill**: **Hill** 斜率, 刻画曲线在 **ec50** 附近的陡峭程度 (绝对值越大越陡)。本参数化下取正值, 对应抑制型 (浓度升高响应下降) 曲线。取值范围 $(0, \infty)$, 拟合时约束在 $[0.05, 20]$ 。

4 NLS 基准拟合

用 base 的 `nls()` 的 port 算法,

```
fit_nls <- function(data) {  
  nls(  
    viability ~ bottom + (top - bottom) / (1 + (conc_um / ec50)^hill),  
    data      = data,  
    start     = list(bottom = -0.2, top = 1, ec50 = 500, hill = 1),  
    lower     = c(bottom = -5, top = 0, ec50 = 1, hill = 0.05),  
    upper     = c(bottom = 1, top = 3, ec50 = 1e5, hill = 20),  
    algorithm = "port",  
    control   = list(maxiter = 1000)  
  )  
}
```

也可以使用 `minpack::nlsLM()`

```
fit_4pl <- function(data) {  
  nlsLM(  
    viability ~ bottom + (top - bottom) / (1 + (conc_um / ec50)^hill),  
    data = data,  
    start = list(bottom = -0.2, top = 1, ec50 = 500, hill = 1),  
    lower = c(bottom = -5, top = 0, ec50 = 1, hill = 0.05),  
    upper = c(bottom = 1, top = 3, ec50 = 1e5, hill = 20),  
    control = nls.lm.control(maxiter = 1000, ftol = 1e-10, ptol = 1e-10)  
  )  
}
```

```
nls_coef <- raw_data |>  
  group_nest(compound) |>  
  mutate(mod = map(data, fit_4pl),  
         coef = map(mod, broom::tidy)) |>  
  select(compound, coef) |>  
  unnest(coef)  
  
nls_coef |>  
  select(compound, term, estimate, std.error) |>  
  knitr::kable(digits = 3) |>  
  kableExtra::kable_styling(font_size = 10)
```

compound	term	estimate	std.error
ent-Paroxol	bottom	-0.158	0.088
ent-Paroxol	top	0.918	0.020
ent-Paroxol	ec50	494.329	71.680
ent-Paroxol	hill	1.529	0.300
3-Fluoro-5-hydroxybenzonitrile	bottom	-0.060	0.050
3-Fluoro-5-hydroxybenzonitrile	top	0.967	0.012
3-Fluoro-5-hydroxybenzonitrile	ec50	545.883	35.726
3-Fluoro-5-hydroxybenzonitrile	hill	2.021	0.314

关键在 **ent-Paroxol** 的 **bottom = -0.158** (负值)，这使它的曲线在高浓度区明显下探、与 3-Fluoro 拉开间距。拟合得到的 IC_{50} 与原文几乎一致 (3-Fluoro-5-hydroxybenzonitrile 约 $546 \mu\text{M}$ 对 $545.9 \mu\text{M}$; ent-Paroxol 约 $500 \mu\text{M}$ 对 $494.4 \mu\text{M}$)，说明数据与模型选择都对得上。

4.1 生成平滑拟合曲线

在对数浓度范围内取密网格，对每个模型 **predict**，得到画线用的数据。

```
library(marginaleffects)

grid <- tibble(conc_um = 10^seq(log10(5), log10(2000), length.out = 600))

pred_data <- raw_data |>
  group_nest(compound) |>
  mutate(
    mod = map(data, fit_4pl)
  ) |>
  mutate(pred = map(mod, ~ predictions(.x, newdata = grid)) ) |>
  select(compound, pred) |>
  unnest(pred)
```

4.2 绘图

```
pal <- c(
  "ent-Paroxol" = "#4A98E0",
  "3-Fluoro-5-hydroxybenzonitrile" = "#F47C7C"
)
```

```

summary_data |>
  ggplot(aes(x = conc_um, y = mean, colour = compound)) +
  geom_line(data = pred_data, aes(y = estimate), linewidth = 0.8) +
  geom_errorbar(aes(ymin = mean - sd, ymax = mean + sd),
                width = 0.06, linewidth = 0.6) +
  geom_point(size = 2.6) +
  scale_x_log10(
    breaks = c(10, 100, 1000),
    labels = scales::label_log()
  ) +
  scale_y_continuous(
    breaks = seq(0, 1.2, 0.2),
    labels = scales::label_number(accuracy = 0.1),
    expand = expansion(mult = c(0.01, 0.02))
  ) +
  scale_colour_manual(
    values = c("ent-Paroxol" = "#4A98E0",
               "3-Fluoro-5-hydroxybenzonitrile" = "#F47C7C")
  ) +
  coord_cartesian(xlim = c(4, 2200), ylim = c(0, 1.2)) +
  labs(x = "Concentration/\u00b5M", y = "Cell Viability", colour = NULL) +
  theme_test(base_size = 15) +
  theme(axis.line = element_line(linewidth = 0.7),
        axis.ticks = element_line(linewidth = 0.7),
        axis.ticks.length = unit(4, "pt"),
        legend.position = c(0.02, 0.14),
        legend.justification = c(0, 0.5),
        legend.text = element_text(size = 12))

```

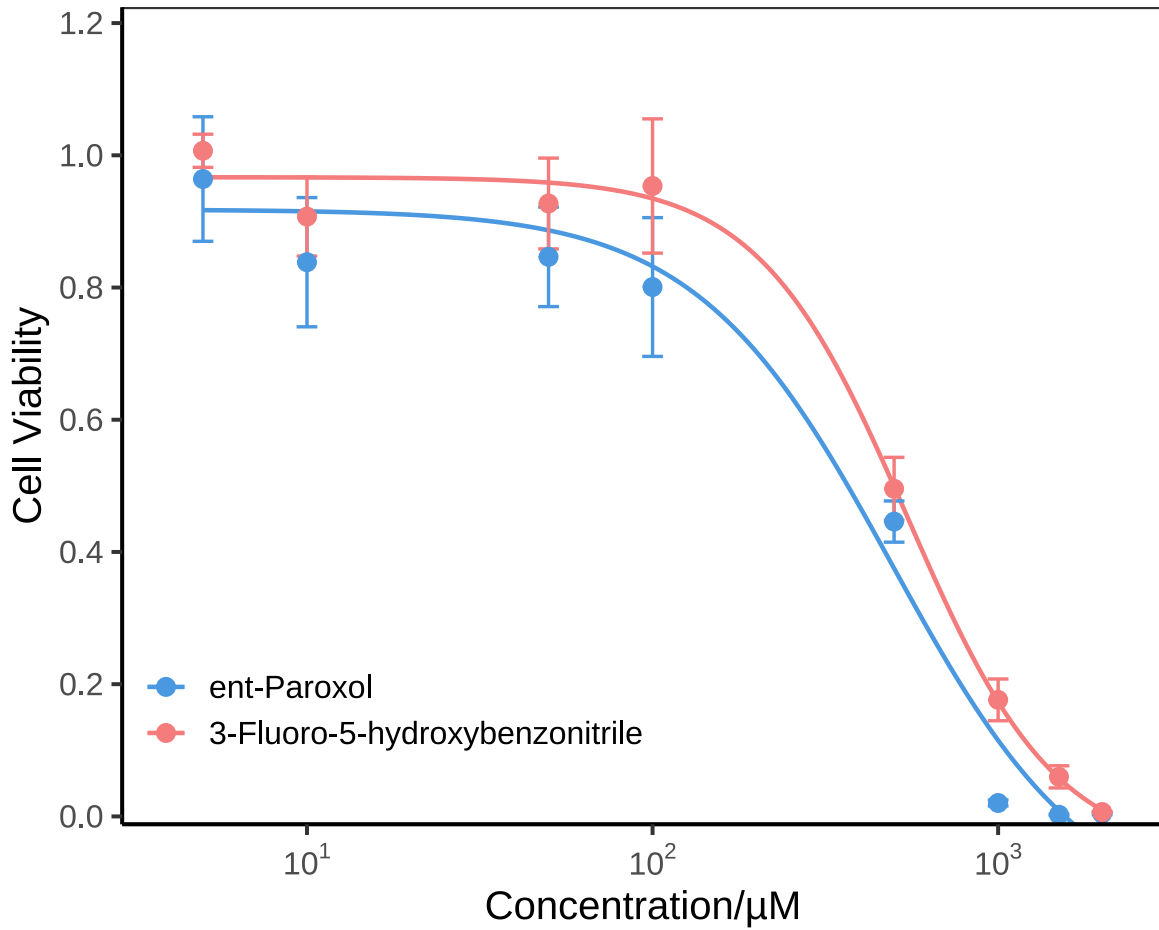


Figure 1: 复现图 4f: Hep G2 细胞存活率对化合物浓度的剂量-反应曲线 (mean $\pm$ sd, n = 6)。

5 用 brms 拟合

5.1 非线性公式

这里，我们把 IC50 放在对数尺度 ($\exp(\text{logic50})$) 保证为正。

```
library(brms)
library(tidybayes)

bform <- bf(
  viability ~ bottom + (top - bottom) / (1 + (conc_um / exp(logic50))^hill),
  bottom + top + logic50 + hill ~ 0 + compound,
  sigma ~ 0 + compound,
  nl = TRUE
)
```

```

bprior <- c(
  prior(normal(0, 0.1), nlpar = "bottom"),
  prior(normal(1, 0.15), nlpar = "top"),
  prior(normal(6.21, 1), nlpar = "logic50"), # log(500) 6.21
  prior(normal(1.5, 0.8), nlpar = "hill", lb = 0)
)

fit <- brm(bform,
  data      = raw_data,
  prior     = bprior,
  backend   = "cmdstanr",
  chains    = 4,
  cores     = 4,
  iter      = 2000,
  seed      = 1024,
  control   = list(adapt_delta = 0.98),
  file      = "fits/brms_fit",
  file_refit = "on_change") # 公式或先验改动后自动重拟合

```

5.2 绘图

后验拟合曲线: 每个化合物在浓度网格上取 `add_epred_draws` 的中位数 + 95% 区间。newdata 必须含模型两个预测变量 (`conc_um`, `compound`)。

```

curve_data <- expand_grid(
  conc_um = 10^seq(log10(5), log10(2000), length.out = 300),
  compound = factor(levels(raw_data$compound), levels = levels(raw_data$compound))
) |>
tidybayes::add_epred_draws(fit) |>
tidybayes::median_qi(.epred, .width = 0.95) |>
ungroup()

curve_data |>
ggplot(aes(colour = compound)) +
  geom_ribbon(
    aes(x = conc_um, ymin = .lower, ymax = .upper, fill = compound),
    alpha = 0.15, colour = NA
  )

```

```

) +
geom_line(
  aes(x = conc_um, y = .epred),
  linewidth = 0.8
) +
geom_errorbar(
  data = summary_data,
  aes(x = conc_um, ymin = mean - sd, ymax = mean + sd),
  width = 0.06, linewidth = 0.6
) +
geom_point(
  data = summary_data,
  aes(x = conc_um, y = mean),
  size = 2.6
) +
scale_x_log10(breaks = c(10, 100, 1000),
              labels = scales::label_math(10^.x)(c(1, 2, 3)),
              limits = c(4, 2200)) +
scale_y_continuous(breaks = seq(0, 1.2, 0.2), limits = c(0, 1.2),
                  expand = expansion(mult = c(0.01, 0.02))) +
scale_colour_manual(values = pal) +
scale_fill_manual(values = pal) +
coord_cartesian(xlim = c(4, 2200), ylim = c(0, 1.2)) + # 只裁剪视图
labs(x = "Concentration/\u00b5M", y = "Cell Viability",
     colour = NULL, fill = NULL) +
theme_classic(base_size = 15) +
theme(axis.line = element_line(linewidth = 0.7),
      axis.ticks = element_line(linewidth = 0.7),
      axis.ticks.length = unit(4, "pt"),
      legend.position = c(0.02, 0.14),
      legend.justification = c(0, 0.5),
      legend.text = element_text(size = 11))

```

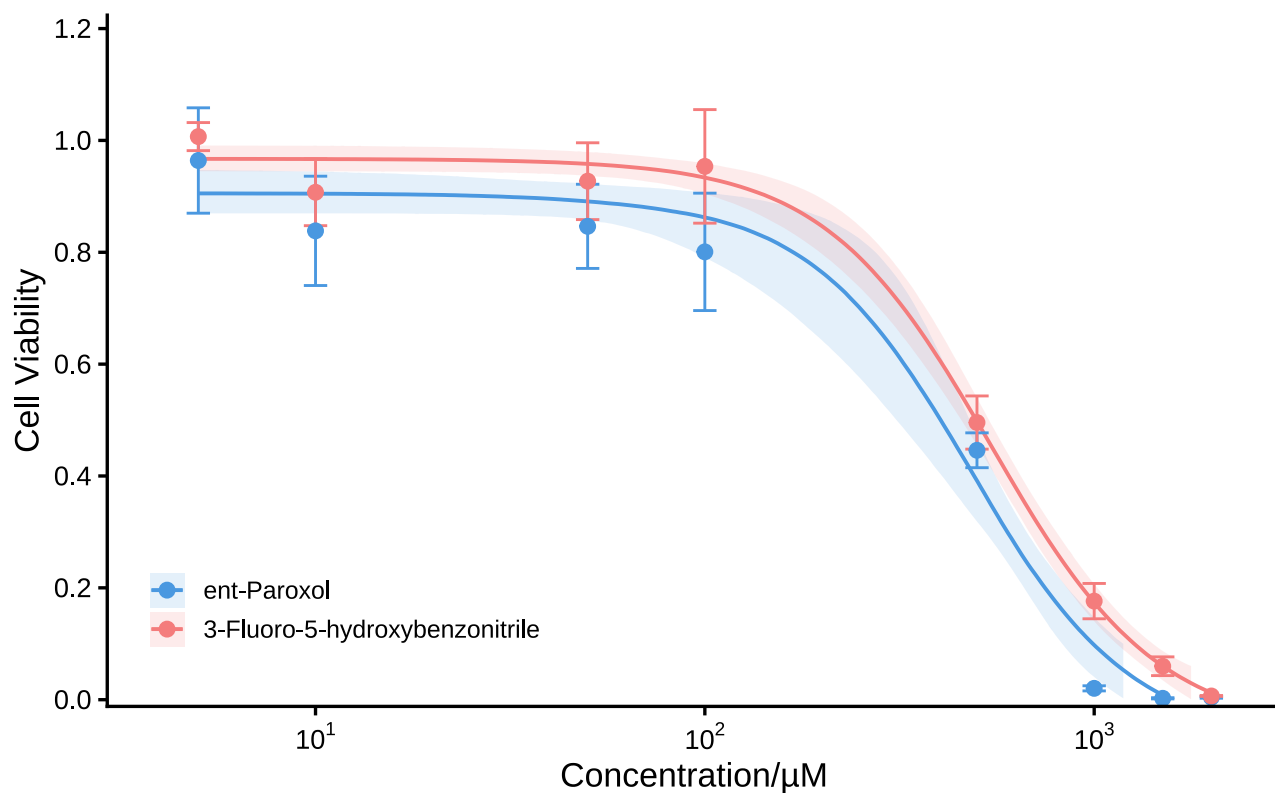


Figure 2: brms 后验中位数曲线

6 用 Stan 拟合

先准备 Stan 需要的数据，包括先验。

```
real<lower=0> prior_bottom_sd; // bottom ~ N(0, prior_bottom_sd)
real<lower=0> prior_gap_sd;    // log_gap ~ N(0, prior_gap_sd)
real prior_hill_mu;            // log_hill ~ N(prior_hill_mu, prior_hill_sd)

dat <- raw_data |>
  mutate(compound_id = as.integer(compound)) |>
  arrange(compound_id, conc_um, replicate)

grid <- crossing(compound_id = seq_along(compound_levels),
                 conc_um = 10^seq(log10(5), log10(2000), length.out = 200)) |>
  mutate(
    compound = factor(compound_levels[compound_id], levels = compound_levels),
    grid_id = row_number()
  )
```

```

stan_data <- list(
  N          = nrow(dat),
  J          = length(compound_levels),
  compound    = dat$compound_id,
  concentration = dat$conc_um,
  y          = dat$viability,
  M          = nrow(grid),
  compound_grid = grid$compound_id,
  concentration_grid = grid$conc_um,

  prior_bottom_sd = 0.2,
  prior_gap_sd    = 0.25,
  prior_hill_mu   = log(1.5),
  prior_hill_sd   = 0.8
)

model <- cmdstan_model("Stan/dose_response_compare.stan", quiet = TRUE)

fit <- model$sample(
  data          = stan_data,
  seed          = 1024,
  chains        = 4,
  parallel_chains = 4,
  iter_warmup   = 1000,
  iter_sampling = 1000,
  adapt_delta   = 0.95,
  max_treedepth = 12,
  refresh       = 0
)

```

Running MCMC with 4 parallel chains...

Chain 1 finished in 2.5 seconds.

Chain 2 finished in 2.7 seconds.

Chain 3 finished in 2.6 seconds.

Chain 4 finished in 2.5 seconds.

All 4 chains finished successfully.

Mean chain execution time: 2.6 seconds.

Total execution time: 2.8 seconds.

```
fit$summary(c("bottom", "top", "ec50", "hill")) |>
  select(variable, mean, median, sd) |>
  knitr::kable(digits = 3)
```

variable	mean	median	sd
bottom[1]	-0.077	-0.055	0.082
bottom[2]	-0.066	-0.061	0.049
top[1]	0.903	0.900	0.023
top[2]	0.968	0.967	0.012
ec50[1]	496.025	496.694	43.295
ec50[2]	549.190	545.795	37.681
hill[1]	3.047	2.540	1.923
hill[2]	2.026	1.998	0.303

7 用 Stan 拟合，放宽先验

把 bottom 的先验从 $\text{normal}(0, 0.2)$ 放宽为 $\text{normal}(0, 1)$ （近似平坦），其余也作调整。目的是想检验先验收缩是不是导致 ent-Paroxol 蓝线被拉高的主因。

```
stan_data <- list(
  N      = nrow(dat),
  J      = length(compound_levels),
  compound = dat$compound_id,
  concentration = dat$conc_um,
  y      = dat$viability,
  M      = nrow(grid),
  compound_grid = grid$compound_id,
  concentration_grid = grid$conc_um,

  prior_bottom_sd = 1,
  prior_gap_sd    = 1,
  prior_hill_mu   = 0,
  prior_hill_sd   = 1.5
)
```

```
fit_wide <- model$sample(
  data      = stan_data,
  seed      = 1024,
  chains    = 4,
  parallel_chains = 4,
  iter_warmup  = 1000,
  iter_sampling = 1000,
  adapt_delta  = 0.95,
  max_treedepth = 12,
  refresh     = 0
)
```

Running MCMC with 4 parallel chains...

Chain 2 finished in 2.7 seconds.

Chain 1 finished in 3.0 seconds.

Chain 4 finished in 2.9 seconds.

Chain 3 finished in 3.0 seconds.

All 4 chains finished successfully.

Mean chain execution time: 2.9 seconds.

Total execution time: 3.2 seconds.

```
fit_wide$summary(c("bottom", "top", "ec50", "hill")) |>
  select(variable, mean, median, sd) |>
  knitr::kable(digits = 3)
```

variable	mean	median	sd
bottom[1]	-0.088	-0.042	0.123
bottom[2]	-0.073	-0.066	0.058
top[1]	0.905	0.901	0.026
top[2]	0.968	0.968	0.012
ec50[1]	510.259	500.090	75.598
ec50[2]	555.159	549.156	43.860
hill[1]	3.787	2.893	3.586
hill[2]	2.010	1.969	0.342

8 参数对比

```
nls_wide <- nls_coef |>
  select(compound, term, estimate) |>
  pivot_wider(names_from = term, values_from = estimate)

stan_median <- function(fitobj) {
  fitobj$summary(c("bottom", "top", "ec50", "hill")) |>
  dplyr::select(variable, median) |>
  mutate(compound_id = as.integer(str_extract(variable, "(?<=\\[\\d\\d)")),
         term = str_remove(variable, "\\[\\d\\d\\]") ) |>
  mutate(compound = factor(compound_levels[compound_id], levels = compound_levels)) |>
  dplyr::select(compound, term, median) |>
  pivot_wider(names_from = term, values_from = median) |>
  select(compound, bottom, top, ec50, hill)
}

bind_rows(
  nls_wide |> mutate(method = "NLS"),
  stan_median(fit) |> mutate(method = "Stan 原先验"),
  stan_median(fit_wide) |> mutate(method = "Stan 放宽 bottom")
) |>
  relocate(method, compound) |>
  knitr::kable(digits = 3)
```

Table 2: NLS 点估计 vs Stan 后验中位数 (ent-Paroxol / 3-Fluoro)

method	compound	bottom	top	ec50	hill
NLS	ent-Paroxol	-0.158	0.918	494.329	1.529
NLS	3-Fluoro-5-hydroxybenzonitrile	-0.060	0.967	545.883	2.021
Stan 原先验	ent-Paroxol	-0.055	0.900	496.694	2.540
Stan 原先验	3-Fluoro-5-hydroxybenzonitrile	-0.061	0.967	545.795	1.998
Stan 放宽 bottom	ent-Paroxol	-0.042	0.901	500.090	2.893
Stan 放宽 bottom	3-Fluoro-5-hydroxybenzonitrile	-0.066	0.968	549.156	1.969

观察: NLS 里 ent-Paroxol 的 `bottom` = -0.158, Stan 后验中位数只有约 -0.05 至 -0.07; 放宽 bottom 先验后恢复到约 -0.07, 仍与 NLS 有明显差异。同时 ent-Paroxol 的 `hill` 从 NLS 的 1.53 变成后验的 2.3-2.6, 且后验很宽 (sd 约 1.8), 说明该参数弱识别。

8.1 曲线间距对比

两条拟合线（ent-Paroxol 蓝、3-Fluoro 红）在网格点上的拟合值之差，取若干浓度点的绝对值。

```
nls_curve <- map_dfr(compound_levels, function(cc) {  
  m <- fit_4pl(filter(raw_data, compound == cc))  
  g <- grid |>  
    filter(compound_id == which(compound_levels == cc)) |>  
    mutate(compound = factor(cc, levels = compound_levels))  
  g |> mutate(fit = predict(m, newdata = g))  
})
```

```
post_curve <- function(fitobj) {  
  mu <- fitobj$draws("mu_grid")  
  med <- apply(mu, 3, median)  
  grid |> mutate(median_fit = as.numeric(med))  
}
```

```
cur_tight <- post_curve(fit)  
cur_wide <- post_curve(fit_wide)
```

网格为对数等距，取最接近目标浓度的点

```
targets <- c(100, 500, 1000, 1500, 2000)  
nn <- sapply(targets, function(t) which.min(abs(grid$conc_um[1:200] - t)))  
  
gap_nls <- function(i) abs(nls_curve$fit[i] - nls_curve$fit[i + 200])  
gap_tight <- function(i) abs(cur_tight$median_fit[i] - cur_tight$median_fit[i + 200])  
gap_wide <- function(i) abs(cur_wide$median_fit[i] - cur_wide$median_fit[i + 200])  
  
tibble(  
  target_conc = targets,  
  grid_conc = grid$conc_um[nn],  
  NLS = sapply(nn, gap_nls),  
  Stan_原先验 = sapply(nn, gap_tight),  
  Stan_放宽bottom = sapply(nn, gap_wide)  
) |>  
  knitr::kable(digits = 3)
```

Table 3: 两条拟合线的间距（拟合值之差，绝对值）

target_conc	grid_conc	NLS	Stan_ 原先验	Stan_ 放宽 bottom
100	98.506	0.102	0.061	0.059
500	500.667	0.124	0.089	0.088
1000	1000.667	0.058	0.094	0.104
1500	1480.039	0.050	0.056	0.055
2000	2000.000	0.054	0.033	0.029

放宽先验能拉开一点，但不足以完全恢复 NLS 的间距（例如 100 μ M 处 NLS 0.102 $\rightarrow$ 放宽后 0.064）。

8.2 曲线叠加图

```
allcurves <- bind_rows(
  nls_curve |> mutate(method = "NLS"),
  cur_tight |> rename(fit = median_fit) |> mutate(method = "Stan 原先验"),
  cur_wide |> rename(fit = median_fit) |> mutate(method = "Stan 放宽 bottom")
)

allcurves |>
  ggplot(aes(conc_um, fit, colour = compound, linetype = method)) +
  geom_line(linewidth = 0.8) +
  geom_point(data = summary_data, aes(conc_um, mean), inherit.aes = FALSE) +
  scale_x_log10() +
  scale_colour_manual(values = pal) +
  labs(x = "Concentration ( $\mu$ M)", y = "Viability",
       colour = NULL, linetype = "拟合方法") +
  theme_bw(base_size = 13)
```

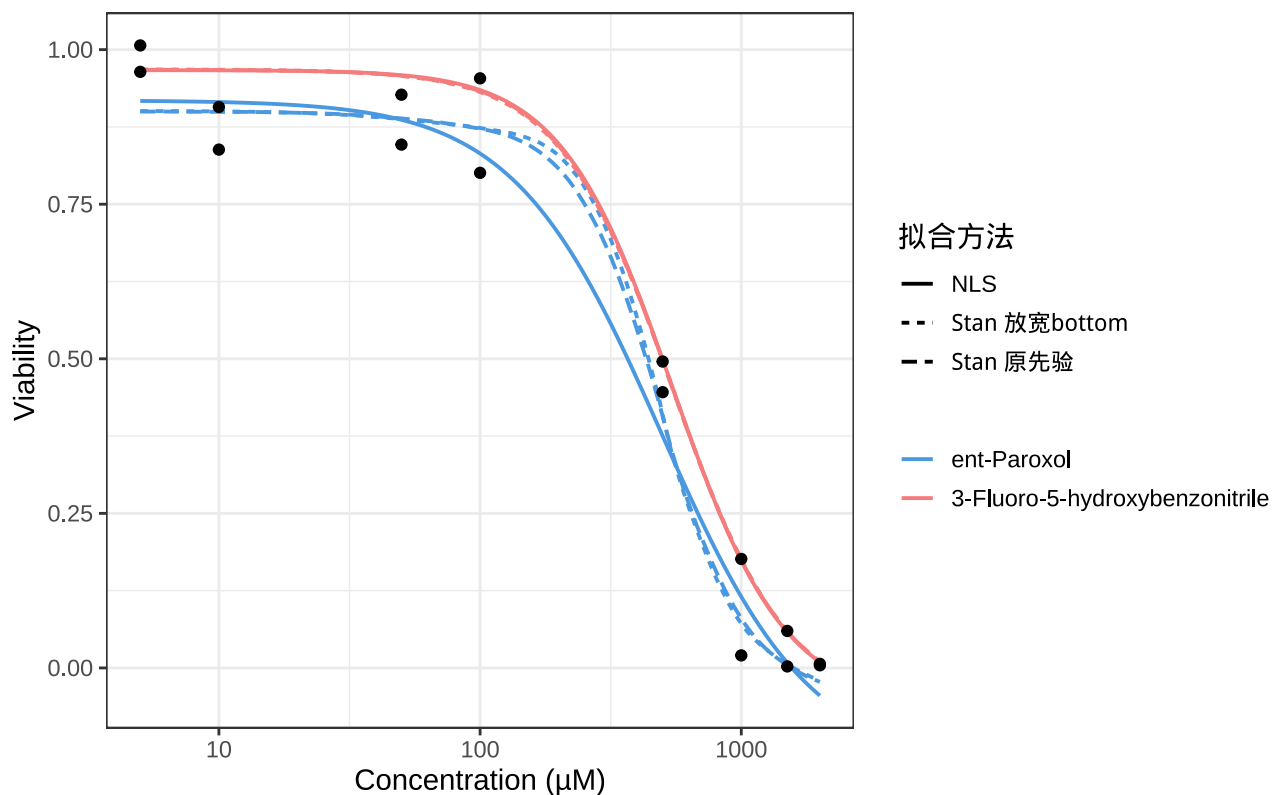


Figure 3: NLS 与 Stan 后验中位数曲线的叠加对比

8.3 收敛诊断

```
fit$summary(c("bottom", "top", "ec50", "hill", "sigma")) |>
  select(variable, mean, median, rhat, ess_bulk) |>
  knitr::kable(digits = 3)
```

Table 4: Stan 拟合的收敛诊断（原先验）

variable	mean	median	rhat	ess_bulk
bottom[1]	-0.077	-0.055	1.004	1026.218
bottom[2]	-0.066	-0.061	1.001	1404.683
top[1]	0.903	0.900	1.002	1674.636
top[2]	0.968	0.967	1.000	3420.312
ec50[1]	496.025	496.694	1.005	2732.658
ec50[2]	549.190	545.795	1.000	1920.703
hill[1]	3.047	2.540	1.005	927.019
hill[2]	2.026	1.998	1.002	1557.849
sigma[1]	0.088	0.087	1.003	2867.844

Table 4: Stan 拟合的收敛诊断（原先验）

variable	mean	median	rhat	ess_bulk
sigma[2]	0.058	0.057	1.001	2902.322

$R^{hat} \approx 1.00$ 、ESS 充足，说明链已收敛——因此差异不是采样/初始值问题，而是先验 + 后验本身。

8.4 小结

放宽先验只能部分恢复间距。`bottom` 是弱识别的外推参数，数据并不能强区分 `bottom = -0.158` 与 `bottom = -0.05`。因此：

- 若目标是**严格复现原图**：原图几乎肯定是 NLS 拟合，保留 `01_run_nls_model.R`；
- 若目标是 **Bayesian 分析**：放宽先验后接受曲线与 NLS 的差异是合理的，用后验区间呈现不确定性，不必强求与 NLS 完全重合。